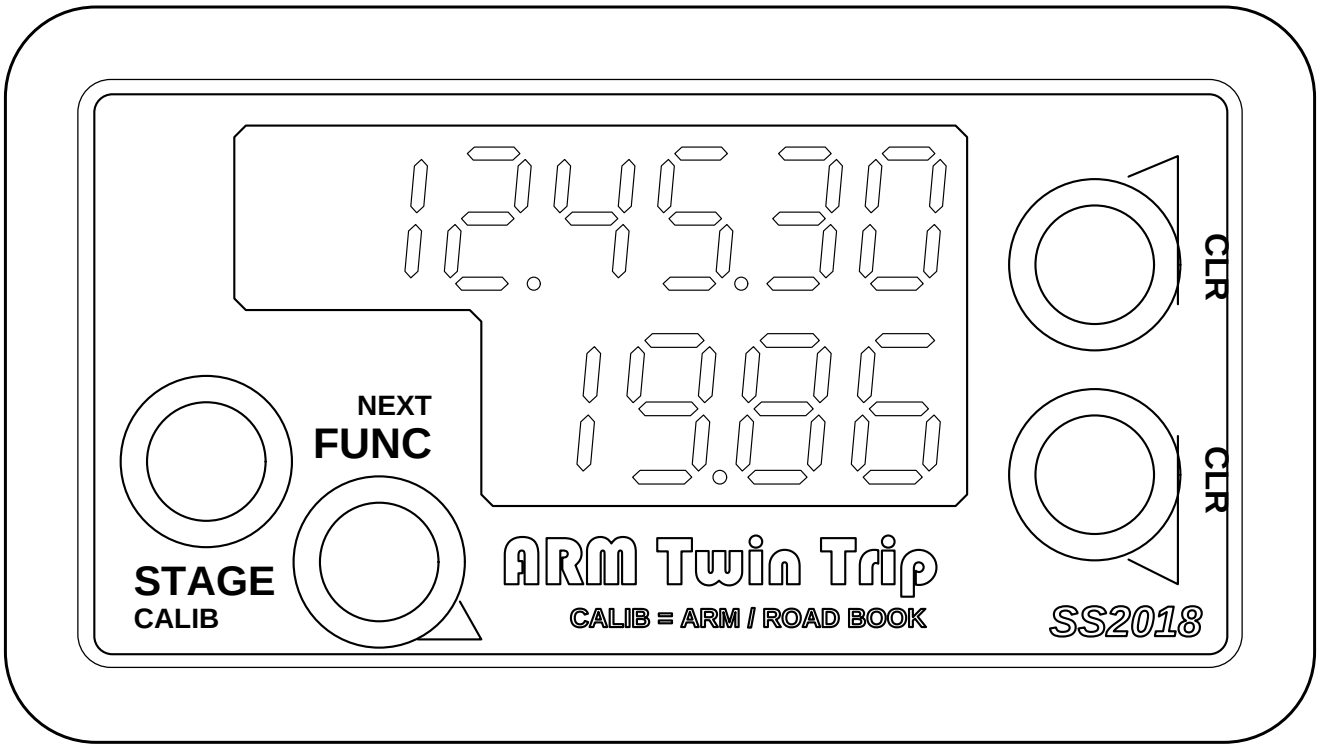
ARM Twin Trip

SS2018

User Manual

Please read this manual before use and keep it for future reference.



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CHAPTER 01

Safety precautions

The ARM Twin Trip SS2018 is a tripmeter designed specifically for stage rallies. Based on the experience of ARM products and many co-drivers, it focuses on the functions needed for stage rallies, combining simple operation with an affordable price.

▲ Power supply
This product is for 12 V DC vehicles only. It cannot be installed in vehicles with other voltages (e.g. 24 V).

▲ Fuse
Always fit a fuse between the vehicle power supply and the unit.

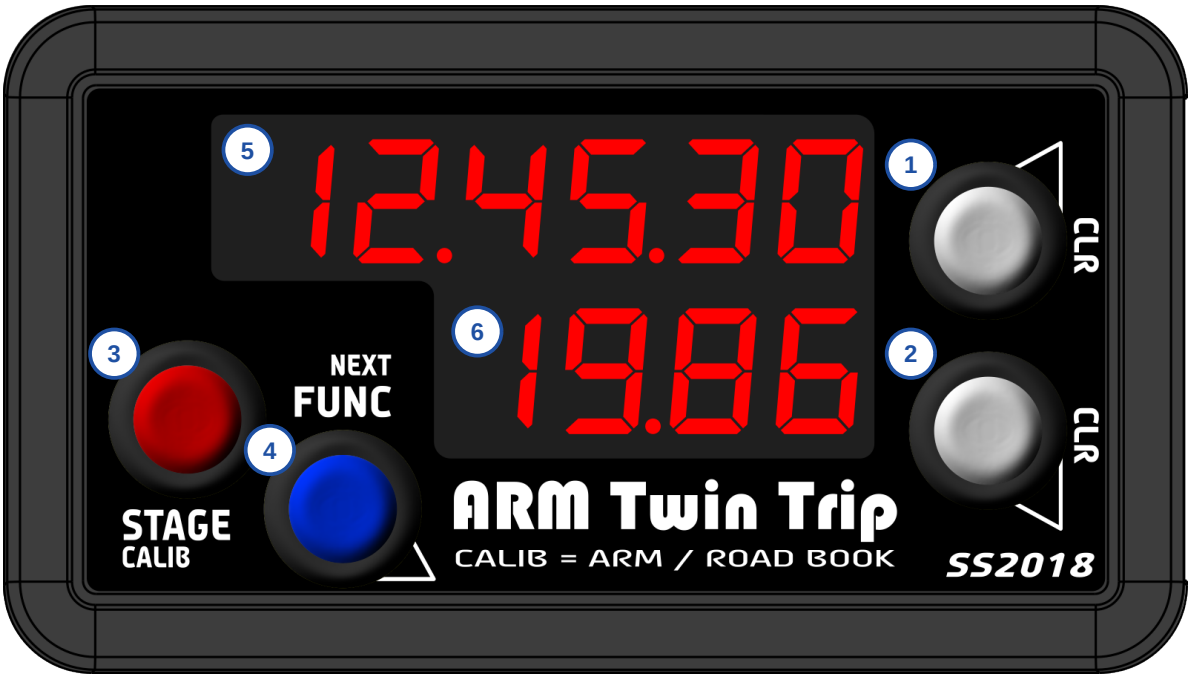
▲ Mounting location
Do not install the unit where it obstructs driving or where it does not comply with regulations, and fix it securely. Do not install it where it can get hot or wet.

Wiring
Bundle the unit’s wiring with tape or cable ties and fix it securely to prevent short circuits or broken wires.

Disassembly and modification
This unit is a precision instrument. Disassembly or modification will stop it from working correctly.

CHAPTER 02

Parts



No.	Name	Function
1	UP button (upper CLR)	Resets the sub trip, holds the time, increases the value while setting. Long press on the clock screen: clock adjust.
2	DOWN button (lower CLR)	Resets the main trip, decreases the value while setting.
3	STAGE button (red, CALIB)	Short press: stage timing. Long press: distance calibration.
4	FUNC button (blue, NEXT)	Short press: switches the upper display; next digit or next setting while setting. Long press: settings (brightness, pulses, unit).
5	Upper display	Clock, sub trip or stage record time.
6	Lower display	Main trip (10 m steps). Stage number on the stage record screen.

Right side: from the top, power switch (slide switch, down = ON), connector B (external switches), connector A (power and speed signal).
Rear: threaded mounting holes (chapter 9).

CHAPTER 03

Connecting to the vehicle

> **▲ Caution**
> Always fit a fuse between the vehicle power supply and the unit.

Connector	Wire	Use
A (required)	Red	+12 V
A (required)	Black	GND (vehicle ground)
A (required)	Pink	Speed signal
B (optional)	Green (2 wires)	Main trip reset (same as the DOWN button)
B (optional)	White (2 wires)	Stage (same as the STAGE button)

Connector A must be connected. Connector B can be left unconnected if you do not use external switches.

Wiring

Connect GND to a point that makes solid contact with the vehicle body. If the point is painted, remove the paint first to avoid a poor contact.
Use momentary switches (ON only while pressed) and connect the two wires of the same color (green or white) to the two terminals of the switch. External switches are sold separately.

CHAPTER 04

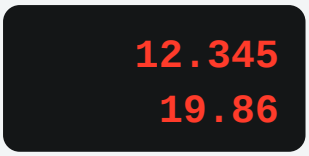
Power and switching the display

Power

The slide switch on the right side is the power switch. Slide it down for ON; the unit starts and shows the clock on the upper display. Slide it up for OFF.

Switching the upper display

Each short press of the FUNC button switches the upper display clock → sub trip → stage records → clock … The lower display shows the main trip.

Screen	Upper display	Example
Clock	Current time (h.mm.ss)	
Sub trip	A distance separate from the main trip (1 m or 10 m steps)	
Stage records	Stage time. The lower display shows the stage number	

About the clock
The clock keeps running when the power is off, but it may drift by about ±1 second per day. Set the time regularly (chapter 8).

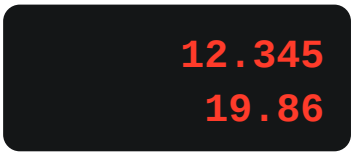
What is kept
Calibration, brightness, pulses and the sub trip unit are kept when the power is off. Stage records are lost when the power is off.

CHAPTER 05

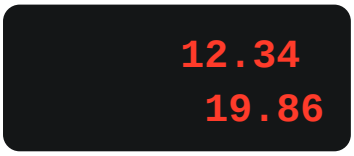
Main trip and sub trip

	Main trip	Sub trip
Display	Lower display (10 m steps)	Upper display (1 m or 10 m steps, selectable)
Reset	DOWN button	UP button on the sub trip screen
Automatic reset	At the stage start and finish	At the stage start

Because the sub trip returns to 0 at the stage start, it can also be used as the total distance from the stage start.
The unit counts forward only.



Sub trip (1 m steps)



Sub trip (10 m steps)

CHAPTER 06

Distance calibration

When the displayed distance does not match the road book, adjust it with the calibration factor. The factor is kept when the power is off.

How to calculate
Calibration factor = ARM displayed distance ÷ road book distance

Reset the factor to 1.0000 before measuring the distance you use to calculate it.

Setting the factor

1. Long press the STAGE button. The factor (C1.0000) appears on the upper display and the digit you can change blinks.
2. Change the value with the UP / DOWN buttons and move to the next digit with a short press of the FUNC button (0.0001–2.9999).
3. Press the STAGE button briefly to finish.

Resetting the factor to 1.0000

1. Long press the STAGE button to show the factor.
2. Long press the STAGE button again. The factor returns to 1.0000.



Calibration screen



Set to 1.0200

CHAPTER 07

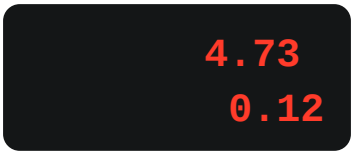
Stage timing

A stopwatch dedicated to stage rallies. It records the times of 30 stages (they are lost when the power is off).

1. Stop at the start line and press the STAGE button briefly.
2. A countdown to 00 s of the next minute starts.
3. The stopwatch starts when the countdown reaches 0. Even during the countdown, it starts automatically as soon as the car starts moving.
4. Five seconds after the start, the upper display switches automatically to the sub trip (distance from the start).
5. Press the STAGE button briefly as you cross the finish line. The stage time is held.
6. After checking it, press the STAGE or UP button briefly to show the time you crossed the finish line.
7. Press the STAGE button again to return to the previous screen, or the UP button to return to the clock. Stage timing ends.



Countdown



Running (sub trip)



Finish

When	Distances reset automatically
Start (STAGE button)	Main trip and sub trip
Finish (STAGE button)	Main trip only

CHAPTER 08

Settings (clock, brightness, pulses, unit)

Setting the clock

1 Switch the upper display to the clock and long press the UP button.

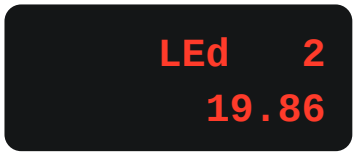
2 Change the blinking value with the UP / DOWN buttons; the FUNC button moves to the next digit.

3 When the reference time reaches the time you set, press the STAGE button. The clock is set.

Brightness, pulses and sub trip unit

Long press the FUNC button to enter the settings; each short press of the FUNC button moves to the next setting. Pressing FUNC on the last setting returns to the previous screen.

Order	Upper display	Setting	Operation
1	LEd	Brightness (3 levels, 0 to 2)	UP: brighter DOWN: dimmer
2	PULS	Speed pulses (2, 4, 8). Set to match the vehicle.	UP / DOWN: switch
3	Unit	Sub trip unit (1 m or 10 m)	UP / DOWN: switch



Brightness



Pulses



Unit

The settings are kept when the power is off.

CHAPTER 09

Mounting

Fix the unit to the car securely, using the threaded holes on the rear or double-sided tape.

Item	Details
Bolt	M4
Bolt length	Thickness of the base + 5 to 7 mm
Hole spacing	48.5 mm, matching the hole spacing of the RAM Mount RAM-B-238U base.
With RAM-B-238U	M4 × 10 mm or M4 × 12 mm bolts are recommended.

▲ Caution
Bolts longer than specified will damage the inside of the unit. Always use bolts of “base thickness + 5 to 7 mm”.

CHAPTER 10

Specifications

Item	Details
Product	ARM Twin Trip SS2018 (model: ARMTPSS2018)
Supply voltage	10–15 V DC (12 V DC vehicles only)
Current consumption	Max. approx. 500 mA
External inputs	2 (main trip reset, stage; switches sold separately)
Operating temperature	−10 °C to 50 °C
Dimensions	W 135 × H 76 × D 27 mm
Distance	Measured from the speed signal. Pulses 2, 4 or 8; calibration factor 0.0001–2.9999
Contents	Unit, user manual, warranty card

CHAPTER 11

Warranty and after-sales service

1. The product comes with a warranty card. The warranty becomes valid when the manufacturer enters the shipping date. Fill in your details and keep the card in a safe place. Warranty cards cannot be reissued.
2. The warranty period is 6 months from the shipping date on the warranty card.
3. Within the warranty period, repairs are free of charge under the warranty terms. Repairs not covered by the terms are charged, even within the warranty period.
4. After the warranty period, the product is repaired for a fee if it can be repaired.
5. Shipping costs from the customer to the manufacturer and the cost of installation and removal are paid by the customer. The manufacturer pays the return shipping after repair.
6. If the product turns out to be beyond repair on arrival, the shipping costs are handled in the same way.
7. The manufacturer does not provide information about the vehicles the product is installed in.

Manufacturer and contact

ARM products URL: https://www.rally-arm.com E-mail: info@rallymonitor.com

CHAPTER 12

Quick reference

Button	Short press	Long press
UP (upper CLR)	Reset the sub trip (sub trip screen) / hold the time / back to the clock after a stage	Clock adjust (clock screen)
DOWN (lower CLR)	Reset the main trip	—
STAGE (red)	Countdown → finish → finish time → end	Calibration (again during calibration: back to 1.0000)
FUNC (blue)	Switch the upper display (clock → sub trip → stage records)	Settings (LEd → PULS → Unit)

Setting screen	UP	DOWN	FUNC	Finish
Calibration (C)	+1	−1	Next digit	STAGE short press
Clock adjust	+1	−1	Next digit	STAGE (at the reference time)
LEd / PULS / Unit	+ / switch	− / switch	Next setting	Last FUNC press

An animated manual is also available: manual.rally-arm.com